

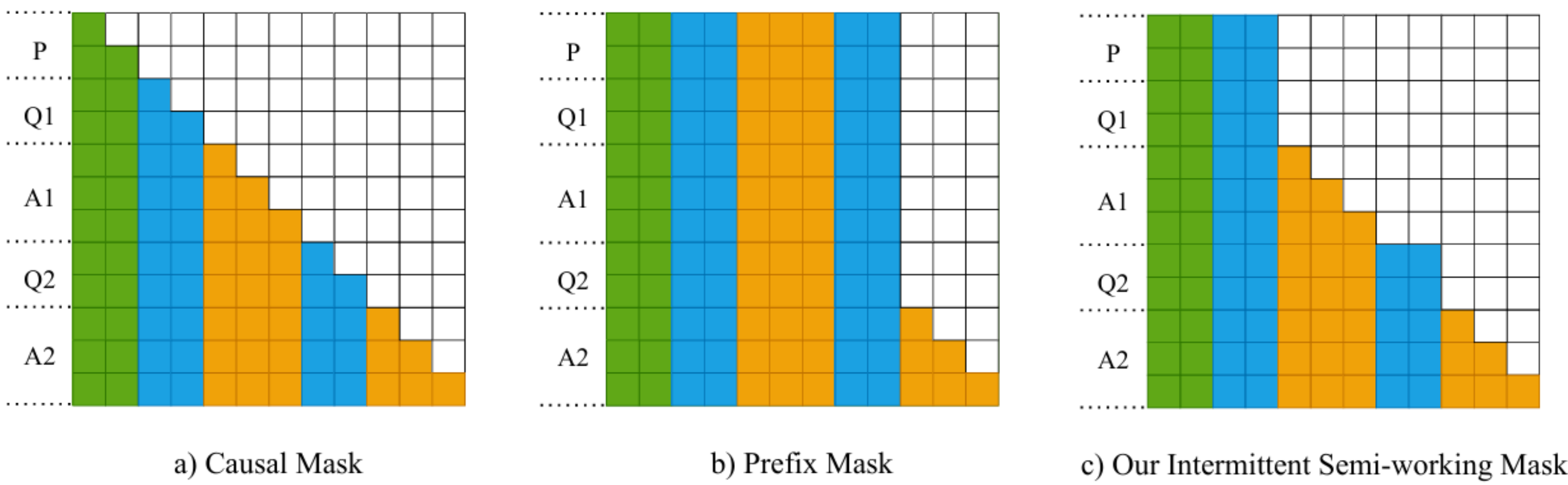


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1. Motivation

- The Problem:** Multi-turn dialogues challenge LLMs to integrate long histories. Standard *Causal Mask* restricts context synthesis, while *Prefix Mask* causes high inference latency (no KV-cache reuse) and requires massive triplet expansion during training.
- Existing Gap:** No solution successfully retains the contextual understanding of prefix mask while preserving the inference-time efficiency of causal mask.
- Our Solution: Intermittent Semi-working Mask (ISM)**, injecting sparse bi-directional attention into the causal backbone.



2. ISM Methodology



Given $X = [P, Q_1, A_1, \dots, Q_{n-1}, A_{n-1}, Q_n]$, ISM defines a boundary function $f(j)$ to control attention scopes:

- Query Segments (Q_k):** Bidirectional attention within the current query and all prior context.
- Answer Segments (A_k):** Strictly unidirectional (causal) attention.
- Future Blocks:** Completely masked out to ensure causality.

3. Mathematical Formulation

Using Linear Self-Attention (LSA) surrogate, ISM modifies the attention bound from j to $f(j)$:

$$x_j \leftarrow x_j + OV \sum_{i=1}^{f(j)} x_i (x_i^\top K^\top Q x_j) \quad (1)$$

Segment Boundary Function $f(j)$:

$$f(j) = \begin{cases} mq_1 & \text{for } 1 \leq j \leq mq_1 \\ mq_k & \text{for } ma_{k-1} < j \leq mq_k, k > 1 \\ j & \text{for } mq_k < j \leq ma_k, k \geq 1 \end{cases}$$

where mq_k and ma_k denote the end positions of query Q_k and answer A_k .

Practical Integration:

- KV-Cache Reuse:** History cache C_h remains valid. Only C_q for the new query is computed and concatenated: $[C_h; C_q]$.
- Zero Overhead:** Architecture-agnostic, parameter-free, drop-in compatible with causal LLMs.

4. Main Results

ISM consistently improves multi-turn dialogue and mathematical reasoning.
A comparison on multi-turn dialogue benchmarks.

Bench	Metrics	LLaMA3.1-8B			Qwen2.5-7B			Qwen2.5-14B		
		Direct	SFT	ISM	Direct	SFT	ISM	Direct	SFT	ISM
MT-Eval	Expansion	3.66	6.21	6.80	4.70	6.63	6.57	6.43	7.34	7.03
	Follow-up	4.23	7.10	7.15	6.03	7.71	7.72	7.55	8.10	8.37
	Recollection	1.39	5.13	5.55	3.84	5.63	5.95	5.32	6.24	6.63
	Refinement	2.09	5.03	5.17	2.92	5.29	5.36	4.31	5.68	5.96
	Average	2.84	5.87	6.17	4.37	6.32	6.40	5.90	6.84	7.00
BotChat	SR($N=16$)	67.3%	77.3%	79.2%	57.6%	71.8%	78.2%	57.0%	85.6%	86.3%
	SR($N=8$)	89.8%	90.5%	93.6%	90.1%	89.4%	92.0%	87.2%	93.4%	94.0%

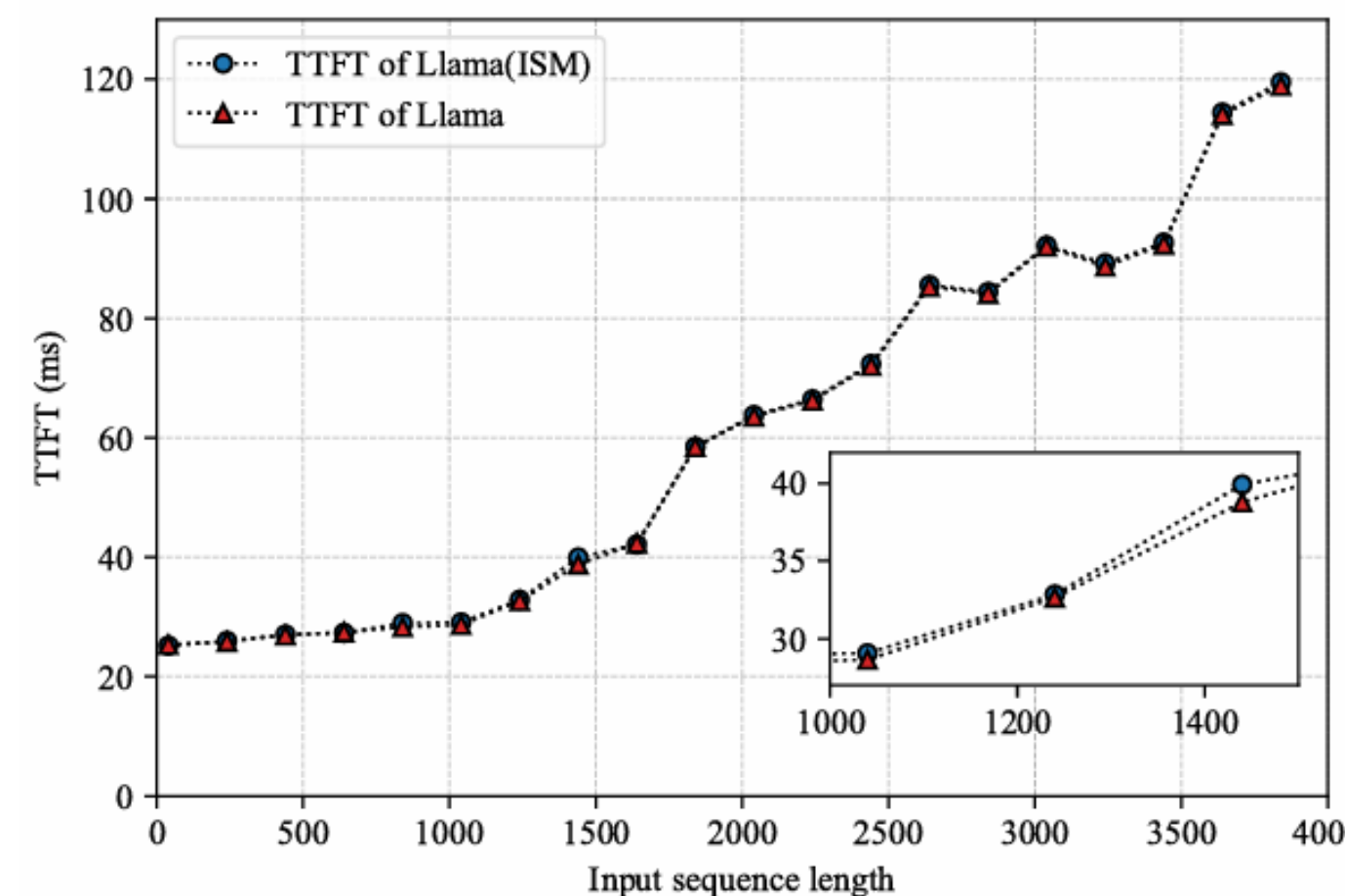
- On **MATH** benchmark, ISM achieves 18.0% accuracy (+2.3% over SFT), proving benefits extend to context-intensive reasoning.
- Largest gains appear in **Recollection**, which heavily depends on integrating prior turns.

5. Latency & Convergence Analysis

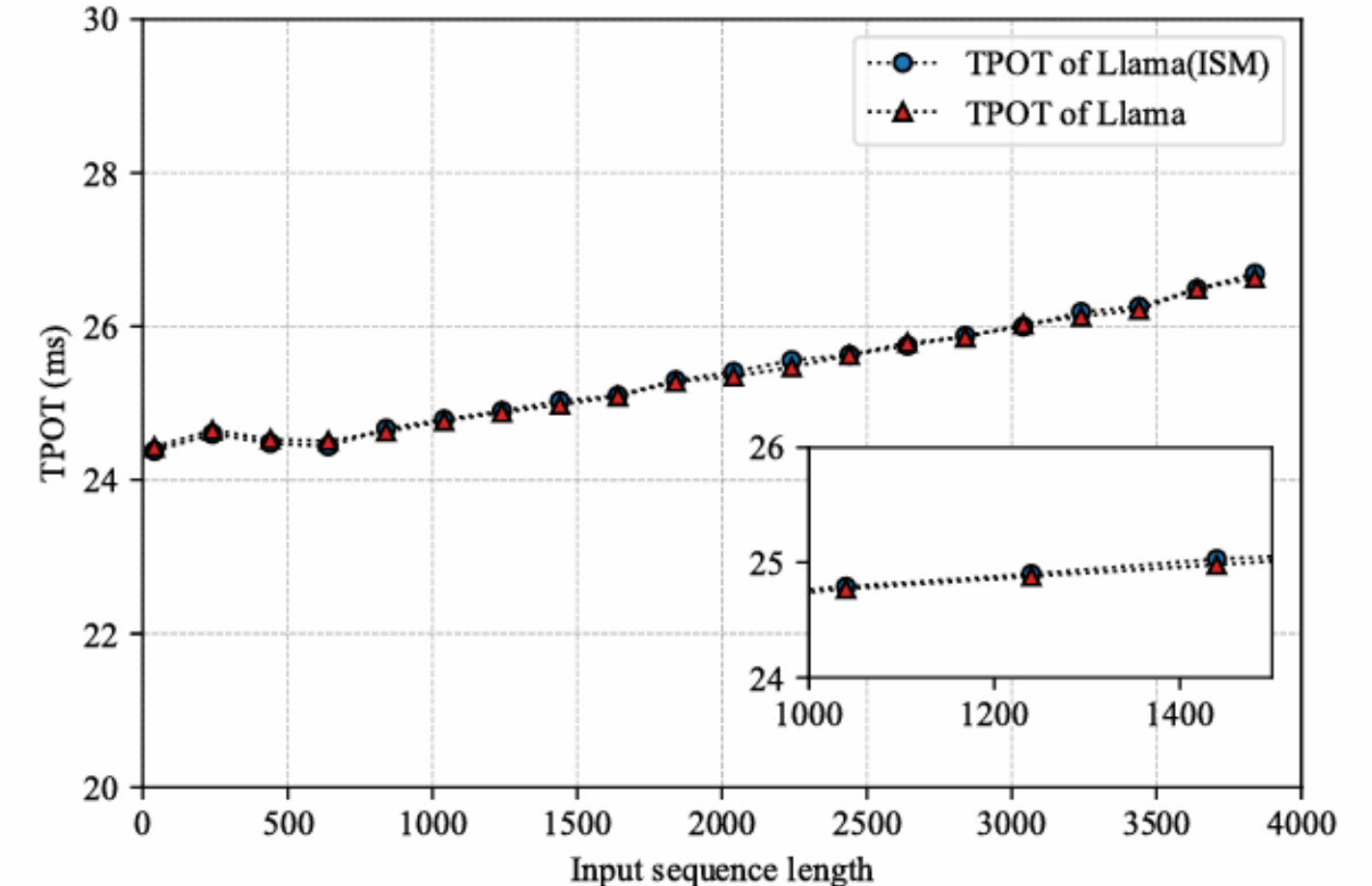
Near-Zero Latency Overhead: Evaluated on 40 simulated dialogues (up to 4,000 tokens) on a single A100 GPU.

- TTFT & TPOT:** ISM closely tracks the SFT baseline. The gap is merely ~ 0.5 –1.0 ms, confirming it inherits causal mask's efficiency.
- Convergence Advantage:** Theoretically, for tokens in the final query window, ISM aggregates all in-context examples, recovering *Batch Gradient Descent* (BGD) dynamics and linear convergence to the optimal w^* , unlike causal mask's suboptimal Online GD.

TTFT and TPOT scale linearly with identical latency.



(a) TTFT latency



(b) TPOT latency

6. Cases

Generated from SFT and ISM (LLaMA3.1-8B), with GPT-4 judgements.

SFT

LLaMA3.1-8B

A: You look rather pale. Are you feeling well?
B: Not very. I was sick most of the night. I didn't sleep very well.
A: I'm sorry to hear that. Maybe you should see a doctor.
B: I think so. I will do that later today.
A: You should take care of yourself. If you don't feel well, you won't be able to pass the test tomorrow.
B: I'm not worried about that. I'll be fine after I get some rest.
A: I hope so.
B: **By the way, what do you think of the weather?**

GPT-4 Judgement

Judgement:
Yes, the conversation is AI involved.
Explanation:
In the 8th chat, B abruptly changes the topic from health and exams to the weather with "By the way, what do you think of the weather?", which is unnatural and seems out of place given the prior context, which is focused on illness and the upcoming test. Human-human conversations usually provide some transition or rationale for such a change, whereas this abrupt switch is more typical in AI-generated exchanges.

ISM

LLaMA3.1-8B

A: You look rather pale. Are you feeling well?
B: Not very. I was sick most of the night. I didn't sleep very well.
A: You look as if you haven't slept for a week.
B: I am usually pretty healthy, but I have been feeling pretty tired lately. I think it's because I've been overworking.
A: You should take it easy for a while.
B: I plan to. I'm going to take a vacation soon.
A: Where are you going?
B: Hawaii. I'm going to spend a week on the beach. I can't wait.

GPT-4 Judgement

Judgement:
No, the conversation is not AI involved.
Explanation:
The conversation flows naturally, with coherent and relevant exchanges between participants. Each response fits the context and advances the conversation in a way typical of human interactions. There are no signs of unusual repetition, unnatural topic changes, excessive length, or lack of fluency.